

The Missing Band Is a Distribution: Do Generative Bandwidth-Extension Models Sample It?

Anonymous submission

Abstract

Generative audio bandwidth extension (BWE) models are presented as samplers of the missing high band, but are evaluated with single-reference metrics that cannot tell a sampler from a regressor. We audit four released 12→48 kHz systems with the checks used for stochastic super-resolution of weather fields (fair CRPS, rank histograms, spread), after showing that when the high band is incoherent with the truth, waveform spread measures output level as much as diversity. FLOWHigh ranks first on LSD and ViSQOL yet is a regressor: one Euler step returns the conditional mean for any start noise. NU-Wave 2 samples, but 8–9 dB too quietly; AudioSR is dispersed but biased by band; every model pulls each utterance’s level toward the corpus mean. On one 88k-parameter network, the deficit is set mostly by the loss domain and is closed by an energy score at no inference cost.

Introduction

Many high bands are consistent with one narrow band, which bounds what any deterministic BWE estimator can recover (Jax and Vary 2002; Bruna, Sprechmann, and LeCun 2016). A point estimate trained on the waveform returns a conditional mean, close to silence for an incoherent band: a muffled output. Diffusion and flow models (Han and Lee 2022; Liu et al. 2024; Yun, Kim, and Lee 2025) are meant to avoid this by sampling $p(y | x)$, yet they are scored by comparing one output with one reference, which rewards an empty band (Su et al. 2021). We found no BWE work that checks whether these models sample. Stochastic down-scaling of weather fields, super-resolution by another name, checks this with proper scores (Gneiting and Raftery 2007) and rank histograms (Leinonen, Nerini, and Berne 2021). We bring those checks to BWE, with (i) a separation of level from dispersion that keeps them meaningful when the band is incoherent, (ii) an audit of four released systems through one scorer, and (iii) a controlled study separating the loss domain from sampling.

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Setup

VCTK 0.92, test split of FLOWHigh and NU-Wave 2: eight speakers unseen in training by every model except possibly AudioSR, with their input protocol (order-8 Chebyshev low-pass, 12 kHz). Levels, LSD and ViSQOL (v3, audio mode) use 240 utterances; calibration uses 32 utterances $\times$ 8 draws. Outputs are scaled to the truth’s energy below 5.5 kHz. The *deficit* is $10 \log_{10}$ of output over true energy per third octave above 6 kHz, averaged over bands. CRPS is the fair ensemble estimator on high-band (HB) log-magnitude; a one-output model scores its MAE. The harness reproduces published LSD/ViSQOL: 0.761/3.61 for FLOWHigh (reported 0.75/3.61), 0.780/3.48 for AP-BWE (Lu et al. 2024) (0.78/3.46); NU-Wave 2 is slightly worse, 0.989/2.57 against 0.94/2.75.

Level Before Dispersion

Let the truth’s HB have energy E , the output’s pE , and let $c = \text{Re}\langle y_H, \hat{y}_H \rangle / E$ be the coherent fraction. With a correct low band, the squared error is

$$\|y - \hat{y}\|^2 = E(1 + p - 2c). \quad (1)$$

An empty band costs E , and only $c > p/2$ beats it, so when $c \approx 0$ silence maximises SNR. On our utterances this ceiling is 22.02 dB at 12 kHz, and a reported SNR S bounds the level: $p \leq 10^{(C-S)/10} - 1 + 2c$. NU-Wave 2’s own table ($C - S = 0.5$ dB) gives at most -8.5 dB; we measure a broadband deficit of -8.9 dB. Every model has $c < 0.02$, including FLOWHigh with attention over the whole utterance, so none recovers the HB waveform.

The same identity governs spread. For M draws independent of one another and of the truth, the gap $10 \log_{10}(\|y - x_i\|^2 / \|y - \bar{x}\|^2)$ is, in expectation, $10 \log_{10} \frac{1+p}{1+p/M}$: a function of level alone. A calibrated sampler has $p = 1$ and a gap of 2.50 dB at $M = 8$. We therefore report level first, and judge diversity by comparing each model’s gap with the independent value *at its own level*.

What the Audit Finds

FLOWHigh is a regressor. It is trained from $x_0 + \epsilon$ (the input’s mel spectrogram plus noise) toward $(x_1 - x_0) - (1 - \sigma)\epsilon$, and it also receives x_0 as conditioning, so at $t = 0$ it can recover

model	deficit	LSD	ViSQOL	gap (indep.)	CRPS
FLowHigh 49M	-1.60	0.761	3.61	0.08 (1.61) [†]	0.682 [†]
AP-BWE 30M	-1.38	0.780	3.48		- 0.777*
NU-Wave 2 1.7M	-7.96	0.989	2.57	0.57 (0.47)	0.583
AudioSR 672M	-0.60	1.561	2.41	2.62 (3.45)	1.243
ours, point 88k	-6.18	0.864	3.13		- 0.889*
ours, sampler 88k	-0.70	0.895	2.83	2.07 (2.09)	0.514

Table 1: One scorer. Deficit (dB), LSD, ViSQOL: 240 utterances. Gap (dB; independent-draw value at the model’s level) and fair CRPS: 32 utterances $\times$ 8 draws. [†]Prior restored. *CRPS = MAE.

arm	objective	deficit	gap	CRPS
det_paper	waveform L1 (LISA)	-21.78	-	2.853*
det	+ log-mag., $\lambda=0.01$	-6.18	-	0.889*
es_marg	energy score	-1.05	2.21	0.537
es_erb	+ ERB bands	-1.08	2.03	0.512
es_erb	$\lambda=0.001$	-5.73	1.16	0.683
es_dec_erb	+ decoder noise, $\lambda=0.1$	-0.70	2.07	0.514

Table 2: One 88k network, identical batches; columns as in Table 1.

ϵ . The optimal field then returns $E[x_1 | x_0] + \sigma\epsilon$ after one Euler step, for any start-noise scale s ; the released script also resets s to 10^{-4} . With the trained prior restored, its draws remain almost identical: the gap is 0.08 dB against 1.61 for independent draws at its level, and the truth lies below or above all eight draws in 41% and 46% of HB bins. LSD and ViSQOL rank it first, and nothing in them registers that it does not sample.

NU-Wave 2 samples, quietly. Its gap (0.57 dB) matches independent draws at its level (0.47); the draws are as diverse as its level allows. What fails is the level: 9.0 dB under on loud frames and 4.7 dB over on quiet ones. Its small spread (0.094) is a level problem, not a diversity problem.

AudioSR is dispersed but biased by band (+5 dB at 11 kHz, -11 dB at 21 kHz; truth above every draw in 52% of HB bins). AP-BWE is a point estimate by design.

Level regresses per utterance. Regressing each utterance’s deficit on the truth’s HB energy share gives slopes of -0.07 (FLowHigh), -0.12 (AP-BWE), -0.32 (NU-Wave 2) and -0.34 (AudioSR); our arms lie between -0.20 and -0.35. Bright utterances come out too dark and dark ones too bright.

What Sets the Deficit

All arms retrain LISA (Kim et al. 2022) on identical batches for 104,950 steps; the samplers add 8 Gaussian input channels, and 4 per output sample in the decoder-noise arm (Table 2). Its released L1 loss leaves the band 21.8 dB short. Adding a log-magnitude term to the same point estimate recovers 15.6 dB; replacing the point loss by a two-draw energy score (Gritsenko et al. 2020; Bouchacourt, Mudigonda, and Nowozin 2016) recovers 5.1 dB more. The loss domain, not point estimation as such, does most of the work: FLowHigh,

a conditional mean in the log-mel domain, is not quiet. The ERB term lowers the band-correlation error from 0.374 to 0.289. Our sampler is not calibrated either: the truth lies above all draws in 21% of HB bins against an ideal 11%, while the lower tail is at its ideal, a residual level bias rather than a narrow ensemble.

Conclusion

Single-reference metrics cannot tell whether a BWE model samples, and when the band is incoherent, spread largely measures level. We suggest reporting level (per band and frame loudness), diversity relative to the model’s own level, and rank tails beside LSD and ViSQOL; CRPS favours samplers over point models by construction, so we compare it among samplers only. Limitations: no listening test, 32 calibration utterances, one seed, VCTK only.

References

- Bouchacourt, D.; Mudigonda, P. K.; and Nowozin, S. 2016. DISCO Nets: Dissimilarity Coefficient Networks. In *Proc. NeurIPS*.
- Bruna, J.; Sprechmann, P.; and LeCun, Y. 2016. Super-Resolution with Deep Convolutional Sufficient Statistics. In *Proc. ICLR*.
- Gneiting, T.; and Raftery, A. E. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477): 359–378.
- Gritsenko, A. A.; Salimans, T.; van den Berg, R.; Snoek, J.; and Kalchbrenner, N. 2020. A Spectral Energy Distance for Parallel Speech Synthesis. In *Proc. NeurIPS*.
- Han, S.; and Lee, J. 2022. NU-Wave 2: A General Neural Audio Upsampling Model for Various Sampling Rates. In *Proc. Interspeech*.
- Jax, P.; and Vary, P. 2002. An Upper Bound on the Quality of Artificial Bandwidth Extension of Narrowband Speech Signals. In *Proc. ICASSP*.
- Kim, J.; Lee, Y.; Hong, S.; and Ok, J. 2022. Learning Continuous Representation of Audio for Arbitrary Scale Super Resolution. In *Proc. ICASSP*.
- Leinonen, J.; Nerini, D.; and Berne, A. 2021. Stochastic Super-Resolution for Downscaling Time-Evolving Atmospheric Fields with a Generative Adversarial Network. *IEEE Transactions on Geoscience and Remote Sensing*, 59(9): 7211–7223.
- Liu, H.; Chen, K.; Tian, Q.; Wang, W.; and Plumbley, M. D. 2024. AudioSR: Versatile Audio Super-Resolution at Scale. In *Proc. ICASSP*.
- Lu, Y.-X.; Ai, Y.; Du, H.-P.; and Ling, Z.-H. 2024. Towards High-Quality and Efficient Speech Bandwidth Extension with Parallel Amplitude and Phase Prediction. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32.
- Su, J.; Wang, Y.; Finkelstein, A.; and Jin, Z. 2021. Bandwidth Extension is All You Need. In *Proc. ICASSP*.
- Yun, J.-H.; Kim, S.-B.; and Lee, S.-W. 2025. FLowHigh: Towards Efficient and High-Quality Audio Super-Resolution with Single-Step Flow Matching. In *Proc. ICASSP*.